The Turning Effect — Paper C

HARD · 50 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Always convert centimetres to metres before you multiply, and give every moment the unit N m.
- The mark for each question is shown in square brackets on the right.
- No formula is given on this paper. If you cannot remember it, write down what you do know and work from there.
- Where a beam has its own weight, that weight acts at its centre.

SECTION 1 — BEAMS WITH WEIGHT

1. A uniform plank is 4 m long and has a weight of 200 N. It is pivoted 1 m from its left-hand end.

(a) State where the weight of the plank acts, and how far this is from the pivot. [2]

(b) Calculate the moment of the plank's weight about the pivot, and state its direction. [2]

2. A uniform beam 3 m long is pivoted exactly at its centre. A 20 N weight hangs 1.2 m to the left of the pivot.

(a) Explain why the weight of the beam itself has no turning effect about this pivot. [2]

(b) A 30 N weight is hung on the right to balance the beam. Calculate its distance from the pivot. [2]

3. Two forces act anticlockwise about a pivot: 5 N at 0.4 m, and 3 N at 0.9 m. A single force is applied 0.5 m from the pivot on the other side to balance them. Calculate its size. [4]

SECTION 2 — THINKING ABOUT THE MODEL

4. A force is applied to the end of a spanner at an angle, rather than at right angles to the handle. Explain why the moment produced is smaller than force × length of handle. [2]

5. A uniform beam 6 m long and of weight 300 N rests on a support at each end. State the upward force provided by each support, and explain your reasoning. [3]

6. A plank 2 m long is found to balance on a pivot placed 0.8 m from its left-hand end. State what this tells you about the plank, and explain how you know. [3]

7. A metre rule of weight 1 N is pivoted at its 40 cm mark. A 2 N weight hangs at the 10 cm mark. The weight of the rule acts at its 50 cm mark.

(a) Calculate the anticlockwise moment produced by the 2 N weight. [1]

(b) Calculate the clockwise moment produced by the weight of the rule itself. [2]

(c) A weight is hung at the 90 cm mark to balance the rule. Calculate its size. [2]

8. Explain how two children of very different weights can still balance a see-saw. [2]

9. A nut requires a moment of 30 N m to loosen it.

(a) Calculate the force needed on a spanner of length 25 cm. [1]

(b) Calculate the force needed on a spanner of length 40 cm. [1]

(c) Comment on what your two answers show. [1]

End of paper. Check every answer has a unit or a form the question actually asked for.